Muhammad Usman Qureshi

Game Engineer · Unity Gameplay, Tools & Tech Art · LiveOps Backend

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Open to relocation (visa sponsorship required) and remote roles

SUMMARY

Game engineer with 5+ years in Unity, from gameplay and editor tools to shaders and VFX. On Hexa Sort (Lion Studios, 10M+ Google Play downloads) I built 10 board mechanics, LiveOps and monetization systems, custom URP shaders and the gameplay for two playable ads, and I built the backend that drives the game's player segmentation. Comfortable across client, server and tech art.

SKILLS

Games: Unity (URP), Cocos Creator, Firebase Remote Config, IAP, AppLovin MAX, AdMob, Adjust, Meta SDK, Addressables, Unity Profiler

Tech art: HLSL/ShaderLab shaders, VFX and particles, lighting, post-processing, Blender

Languages: TypeScript, JavaScript, C#, PHP, SQL, HLSL, Python

Backend: Node.js, Express, PostgreSQL, MySQL, Redis, BullMQ, REST APIs, OAuth 2.0/JWT, Laravel, Snowflake, Docker, Google Cloud

Frontend: React, TanStack Start, Tailwind CSS, shadcn/ui, Zod, Vitest, React Testing Library

AI & workflow: Claude Code, MCP servers, multi-agent development, TDD, spec-driven design, GitHub Actions, Git

EXPERIENCE

Game Developer · Mindstorm Studios, Lahore, Pakistan — Aug 2023 – Present

Developer on Hexa Sort and other mobile titles; since 2025 also building the backend services, SDKs and internal tools behind them.

Unity game client: Hexa Sort, Tile Garden & other titles (2023 – 2025)

- **Built 10 board mechanics for Hexa Sort** (including Drone, Toaster, Kettle and Generator Nuts), time-based levels with a Time Freeze booster, and server-timed boosters with Android and iOS local notifications.
- **Built LiveOps and monetization systems:** an in-game mailbox with news cards targeted by app version, level, platform and country, remote and bulk gifting, IAP and restore flows, rewarded-ad systems, on-device A/B bucketing and Firebase Remote Config kill switches.
- **Wrote custom URP shaders in HLSL** (mobile cel shading, shadow-receiving transparent ground, SRP Batcher compatible) that replaced the third-party shaders on Hexa Sort's board, and extended an SDF rounded-corner tile shader for a jigsaw game.
- **Built editor tools that sped up the team's work:** a code generator that scaffolds a new tile mechanic (scripts, prefabs and config), level-editor features and a batch screenshot uploader for the level library.
- **Built the gameplay for two shipped Hexa Sort playable ads in Cocos Creator (TypeScript)**, each showcasing a new mechanic (Mole, Shoot-to-Sort) with its own input, animation and level setup.
- **Tile Garden (Lion Studios, 1M+ downloads) and legacy titles:** Helpshift support, seasonal event UI and Lion SDK, Google UMP and iOS ATT migration; revived two older coin-pusher games with Unity and Android API upgrades, rebuilt ad mediation and fixed IAP.

Backend, LiveOps platform & tooling (2025 – Present)

- **Designed and built Hexa Sort's player-segmentation service end to end:** a data-driven rule engine (Node.js, TypeScript, Express, PostgreSQL, Redis/BullMQ, Google Cloud SQL) that decides which economy and ad configuration each player receives. Analytics validated 95.8% segment coverage after rollout.
- **Built a secure admin API and web console** (React, TanStack Start, Tailwind, shadcn/ui, Zod) so Product and UA teams author, preview and publish rules themselves: OAuth 2.0 with PKCE, JWT, CSRF protection, rate-limited login and audit logs, plus a rule-impact preview that shows before/after player distribution before a change goes live.
- **Built a remote MCP server** (read-only, OAuth-protected, allow-listed) that lets teammates query, explain and simulate live segmentation rules from Claude; shipped the scoring pipeline and dashboard for an early-payer prediction model.
- **Initial architect and co-developer of the studio's Unity analytics SDK** (UPM package): server-controlled event filtering, IAP receipt validation, ad-revenue, session and FPS tracking; automated releases to a private package registry with GitHub Actions.

- **Integrated a direct-to-consumer web store (Appcharge) into three titles** as a primary technical contact for the rollout, and root-caused cross-SDK attribution and analytics issues with Analytics, UA, QA and vendors (Adjust, Meta).
- **Work spec-first and test-first with AI coding agents (Claude Code):** about 150 written design specs and implementation plans and 1,800+ automated tests across two services; mentored a junior engineer through code review and onboarded three engineers onto the service.

Technical Artist, VFX Artist & Game Developer · Codora (PD Labs), Islamabad, Pakistan May 2022 – Jul 2023

- Built real-time environments for Portal Dwellers in Unity (HDRP), combining lighting, post-processing, shaders and VFX, and rendered in-engine videos for marketing and client deliverables.
- Core team member spanning tech art and programming: render-pipeline setup, lighting, profiling and gameplay features.

Game Developer · AHGamesStudios, Islamabad, Pakistan Nov 2021 – Apr 2022

- Implemented gameplay mechanics and interactive elements for multiple mobile titles and produced in-engine cinematics for a tank-battle game.
- Optimized performance and loading times across devices, and trained an intern.

Junior Game Developer · Alam Digital Gaming, Islamabad, Pakistan Feb 2021 – Oct 2021

- Built vehicle mechanics, game levels and design prototypes, plus tools and scripts that sped up development; diagnosed and fixed performance issues.

ACHIEVEMENTS

- **Most Engaging Game award** at Mindstorm Winter Game Jam 2022 for Makhi Museebat, later taken towards commercial release with Mindstorm Studios.
- **Most Hilarious Gameplay award** at Mindstorm Rookies Game Jam 2022.
- **Mentor** at the PGDC game jam, Air University.

EDUCATION

Bachelor's in Computer Science · PMAS Arid Agriculture University, Rawalpindi 2023